Separable simplicial algebras

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Contents

- Motivations
- Simplicial algebras and simplicial modules
- Simplicial enrichment
- Separable simplicial algebras
- Twisted Calabi-Yau simplicial algebras
- Future questions

Why separable simplicial?

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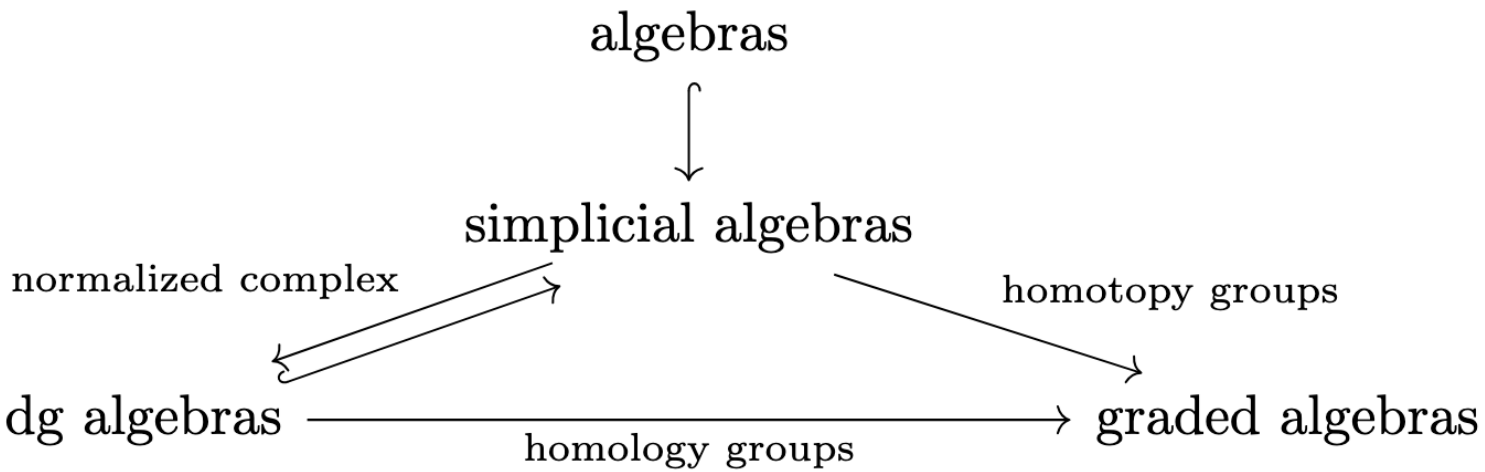
- Twisted Calabi-Yau algebras of dimension 0 $\iff$ separable (Reyes-Rogalski 22').
- Twisted Calabi-Yau algebras are studied in more general settings: differential graded (Ginzburg 07' etc.) $\to$connected graded (Reyes-Rogalski-J.Zhang 14' etc.) $\to$ group-graded (Baki 25' etc.) $\to$ monoidal category (Baki-Reyes 26').
- Separable algebras have been studied in (braided) monoidal categories (Pareigis 76', Van Oystaeyen-Y.Zhang 98') $\to$ (tensor-)triangulated categories (Balmer 11', 16')$\to$ symmetric monoidal ∞-categories (Naumann-Pol-Ramzi 26').

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1. Define and characterize separable simplicial R-algebras.
2. Study related invariants such as the Brauer group.
3. Give a tentative definition of twisted Calabi-Yau simplicial algebras and investigate if twisted Calabi-Yau simplicial algebras of dimension 0 are precisely the separable ones.

Simplicial objects

Definition

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$$d_i d_j = d_{j-1} d_i \quad i < j,$$
$$s_i s_j = s_{j+1} s_i \quad i \leq j.$$

$$d_i s_j = \begin{cases} s_{j-1} d_i & i < j, \\ \mathrm{id} & i = j \text{ or } i = j+1, \\ s_j d_{i-1} & i > j+1. \end{cases}$$

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$$\cdots X_2 \;\underset{\longleftarrow s_0 \longleftarrow}{\overset{d_0 \longrightarrow}{\longleftarrow d_1 \longrightarrow}}\; X_1 \;\underset{d_1}{\overset{d_0}{\longleftarrow s_0 \longrightarrow}}\; X_0$$

Examples of simplicial objects

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The representable functor $\Delta^n = \mathrm{Hom}_{\Delta}(-, [n])$ is a simplicial set and $\Bbbk\Delta^n$ is the simplicial $\Bbbk$-module freely generated by Δ^n.

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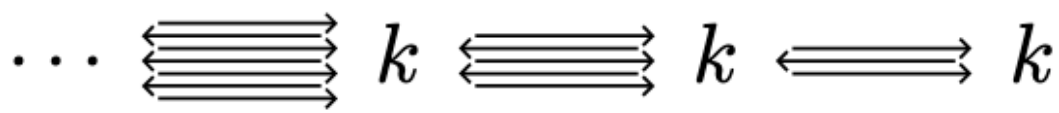
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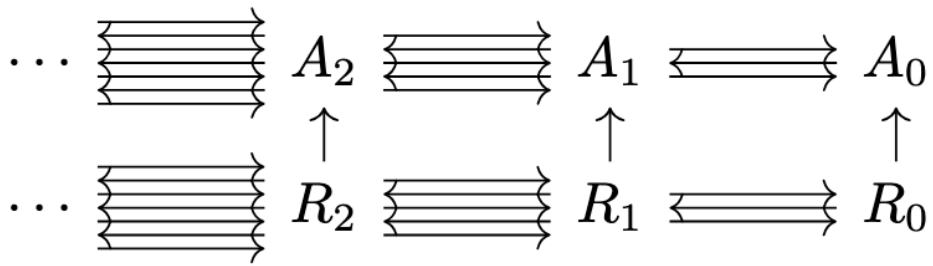
Singular and Hochschild chain complexes are simplicial abelian groups.

Simplicial algebras and simplicial modules

- A simplicial ring R is a simplicial object in Ring. Equivalently, it is a monoid in the category of simplicial abelian groups.

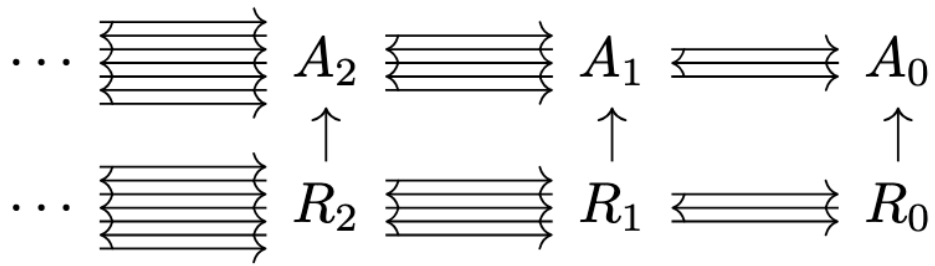
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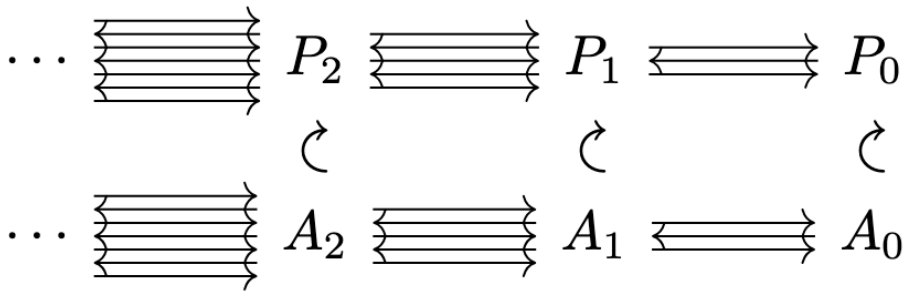


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- Simplicial modules are defined as module objects over monoid and the tensor product is defined degreewise.



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Remark

Simplicial R-algebras A and B are simplicial Morita equivalent if $A\,\mathrm{sMod} \simeq B\,\mathrm{sMod}$ as enriched categories.

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Definition (HSZZ 26')

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Basic projectives fit better than f.g. projectives in simplicial Morita theory.

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Let A be a simplicial R-algebra and $A^e = A \otimes_R A^{op}$. We say A is

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One can also define A to be Azumaya if A^e is simplicial Morita equivalent to R (i.e, $A^e\,\mathrm{sMod} \simeq R\,\mathrm{sMod}$ as enriched categories.)

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Remark

It is not known in general whether central-separable Azumaya monoids agree with Morita-theoretic Azumaya monoids (Pareigis 76', Borceux-Vitale 02'). Our simplicial Auslander-Goldman theorem answers this question for simplicial algebras.

Some rigidity results

Proposition (HSZZ 26')

Let A be a simplicial $\mathbb{k}$-algebra. Then A is separable over $\mathbb{k}$ if and only if A_0 is separable over $\mathbb{k}$ and $A \cong A_0$, the constant simplicial algebra determined by A_0.

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Theorem (HSZZ 26')

Let A be a simplicial R-algebra. Then A is separable over R if and only if $Z(A)$ is separable over R, A_0 is Azumaya over $Z(A)_0$, and $A \cong Z(A) \otimes_{Z(A)_0} A_0$ as simplicial $Z(A)$-algebras.

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Theorem (HSZZ 26')

Let R be a commutative simplicial ring. Then $Br(R) \cong Br(R_0)$.

Twisted Calabi-Yau conditions

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(2)

$$\mathbf{Ext}^i_{A^e}(A, A^e) \cong \begin{cases} 0 & i \neq d \\ A^\mu & i = d \end{cases}$$

as right simplicial A^e-modules for $\mathbf{Ext}^i = H^i(R\mathbf{Hom})$ and some automorphism $\mu : A \to A$.

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There is a resolution of left simplicial A^e-modules

$$0 \to A^e = R[x,y] \xrightarrow{x-y} A^e = R[x,y] \xrightarrow{mult.} A = R[x] \to 0.$$

$A = R[x]$ is a twisted Calabi-Yau simplicial algebra of dimension 1.

Some rigidity results

Lemma

Let P be a basic projective left simplicial A-module. Then

$$A \otimes_{A_0} P_0 \cong P \qquad a \otimes p \mapsto a \cdot s_0^n(p)$$

is an isomorphism of left simplicial A-modules.

Corollary

Suppose there is a basic projective resolution $Q_1 \to Q_0 \to M \to 0$ of left simplicial A-modules. Then $A \otimes_{A_0} M_0 \cong M$ as left simplicial A-modules.

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Proposition

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(3) *If $R = \Bbbk$ is the constant simplicial ring, then $A \cong A_0$ which is the constant simplicial algebra determined by A_0.*

Remark

Over $\Bbbk$, the twisted Calabi-Yau simplicial algebras are no different from ordinary twisted Calabi-Yau algebras. This is consistent with our rigidity result for separable simplicial $\Bbbk$-algebras.

Questions for future studies

(1) Twisted Calabi-Yau of dim 0 vs. separable.

– If A is a twisted Calabi-Yau simplicial R-algebra of dimension 0, it is (basic) projective as a left simplicial A^e-module, whence separable.

– If A is separable, then $\mathbf{Ext}^i_{A^e}(A, A^e) = 0$ for $i > 0$. It remains to show $\mathbf{Hom}_{A^e}(A, A^e) \cong A$.

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(2) Nakayama automorphism.

– A twisted Calabi-Yau simplicial R-algebra A satisfies $A \cong A^e \otimes_{A_0^e} A_0$. Is the Nakayama automorphism of A always extended from that of A_0?

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(3) Basic projective vs. f.g. projective.

– The rigidity of twisted Calabi-Yau simplicial algebras comes from the basic projective resolution in the definition. What changes if the resolution uses arbitrary f.g. projectives?

Thank you!

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